

Supplementary material for *Dynamic consequences of declining vaccination coverage in a heterogeneous world*

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1 Theoretical honeymoon time in the SIR model

[Emily: I've rewritten the first half of this section to define the model equations and derive the equilibria explicitly.] We consider here how reductions in vaccination can lead to rebounds in disease prevalence. In particular, we will focus on the post-vaccination “honeymoon”, or how long it takes for the disease to rebound after vaccination starts declining. We use the term “effective immunization coverage” throughout for the proportion successfully immunized, thereby accounting for both vaccination coverage and imperfect vaccine efficacy.

We begin with an SIR (Susceptible, Infected, Recovered) model of disease transmission. We assume that births and deaths occur at the same rate, μ , and some proportion, v , of births are effectively immunized. We further assume that new infections occur upon contact between susceptible and infected individuals at rate β , and infected individuals recover at rate γ . The model can be defined by the following system of ordinary differential equations:

$$\frac{dS}{dt} = \mu(1 - v) - \beta SI - \mu S \quad (1)$$

$$\frac{dI}{dt} = \beta SI - (\gamma + \mu)I \quad (2)$$

$$\frac{dR}{dt} = \mu v + \gamma I - \mu R \quad (3)$$

where S , I , and R represent the proportions of the population that are susceptible, infected, and recovered, respectively.

One way to understand the dynamics of the SIR model is through its equilibrium behavior, or the conditions under which the system does not change over time. The equilibrium conditions are given by setting the derivatives to zero, and for the I compartment, we have

$$\frac{dI}{dt} = \beta SI - (\gamma + \mu)I = I(\beta S - (\gamma + \mu)) = 0, \quad (4)$$

which demonstrates the existence of two equilibria:

1. disease-free equilibrium, where $I^* = 0$ and $S^* = 1 - v$, and

2. endemic equilibrium, where $I^* = \frac{\mu}{(\gamma+\mu)}((1-v) - 1/\mathcal{R}_0) > 0$ and $S^* = \frac{\gamma+\mu}{\beta} = 1/\mathcal{R}_0$.

This equilibrium point only exists and is positive when $v < 1 - 1/\mathcal{R}_0$. *[Emily: the exact equation for I^* in this case is less relevant for our purposes, but I've included it here for completeness]*

Within this framework, we imagine a population with some “pre-decline” effective immunization coverage, v_0 , that experiences an instantaneous drop in coverage to v . We consider only the disease-free case in which the pre-decline effective immunization coverage is above the herd-immunity threshold, $v_0 > 1 - 1/\mathcal{R}_0$, equivalently $S^* = 1 - v_0 < 1/\mathcal{R}_0$; otherwise there is no positive honeymoon period. *[Emily: flagging that we haven't formally introduced the definition of honeymoon period yet, but perhaps safe to say the reader has been through the main text and has the general idea]*

We hypothesize that the duration of the honeymoon period, or informally, the time until the first rebound outbreak, will depend on a number of factors including the basic reproduction number of the pathogen $\mathcal{R}_0$, the equilibrium proportion susceptible S^* , the birth rate μ , the post-decline effective immunization coverage v , and the timing of importations. Because the timing and establishment of importations cannot be predicted by this deterministic model, we define the theoretical honeymoon time as the time until $\mathcal{R}_{\text{eff}} = \mathcal{R}_0 S(t)$ crosses 1. This provides a lower bound on the actual rebound time, because rebound additionally requires an importation to arrive and establish.

We can calculate analytically the time at which $\mathcal{R}_{\text{eff}}$ crosses 1 under the assumptions of local extinction ($I = 0$), constant post-decline effective immunization coverage v , and a constant population scaled to 1. The proportion susceptible satisfies

$$\frac{dS}{dt} = \mu(1 - v) - \mu S = \mu[(1 - v) - S]. \quad (5)$$

Because the population is initially at the disease-free equilibrium associated with the pre-decline effective immunization coverage v_0 , we have $S(0) = 1 - v_0$. Since $v < v_0$, we have $S(0) < 1 - v$, so Equation (5) shows that $S(t)$ increases toward $1 - v$. Thus we have a first-order linear initial-value problem, whose solution is

$$S(t) = 1 - v - (v_0 - v)e^{-\mu t}. \quad (6)$$

The critical effective immunization coverage (corresponding to the herd-immunity threshold) is

$$v_{\text{crit}} = 1 - \frac{1}{\mathcal{R}_0}. \quad (7)$$

If effective immunization coverage declines from above to below this level, so that

$$v < v_{\text{crit}} < v_0, \quad (8)$$

then $S(t)$ increases from $1 - v_0$ and crosses $1/\mathcal{R}_0 = 1 - v_{\text{crit}}$ after a finite time. Setting $\mathcal{R}_{\text{eff}}(t) = \mathcal{R}_0 S(t) = 1$ gives the theoretical honeymoon time,

$$T = \frac{1}{\mu} \log \left(\frac{v_0 - v}{v_{\text{crit}} - v} \right) = \frac{1}{\mu} \log \left(\frac{v_0 - v}{1 - 1/\mathcal{R}_0 - v} \right). \quad (9)$$

This expression is positive because $v < v_{\text{crit}} < v_0$ implies $(v_0 - v)/(v_{\text{crit}} - v) > 1$. If $v \geq v_{\text{crit}}$, $\mathcal{R}_{\text{eff}}$ never exceeds 1 and the theoretical honeymoon time is infinite; if $v_0 \leq v_{\text{crit}}$,

the population is not initially above the herd-immunity threshold and there is no positive honeymoon period of this kind.

Equivalently, the pre- and post-decline values of effective immunization coverage that yield the same honeymoon time T satisfy

$$v_0 = v + e^{\mu T}(v_{\text{crit}} - v) = v(1 - e^{\mu T}) + e^{\mu T}v_{\text{crit}}. \quad (10)$$

For fixed T and μ , Equation (10) defines a straight line in the plane whose horizontal and vertical coordinates are the post- and pre-decline effective immunization coverages, v and v_0 , respectively. The line has slope $1 - e^{\mu T} < 0$ and vertical intercept $e^{\mu T}v_{\text{crit}}$. Consequently, combinations of v and v_0 that produce the same theoretical honeymoon time lie along straight lines. This explains the linear contours in the heatmap of theoretical honeymoon time (main text, Figure 2B).

2 Age-structured model

[Emily: This section has been updated and all code/figures have been rerun. Minor numerical differences and no qualitative changes.]

We also test honeymoon times under various assumptions about age-structured contact patterns. Our age-structured model defines a discrete set of A age classes. Transmission between individuals of different age classes is governed by a who-acquires-infection-from-whom (WAIFW) matrix, $\theta_{i,j}$, where i and j index age classes. The full system of ordinary differential equations governing the proportion of individuals in each age class and disease state is as follows:

$$\frac{dS_{i=1}}{dt} = \mu(1 - v) - \sum_{j=1}^A \theta_{1,j}S_1I_j - \mu S_1 - \alpha_1 S_1 \quad (11)$$

$$\frac{dS_{i \neq 1}}{dt} = \alpha_{i-1}S_{i-1} - \sum_{j=1}^A \theta_{i,j}S_iI_j - \mu S_i - \alpha_i S_i \quad (12)$$

$$\frac{dI_{i=1}}{dt} = \sum_{j=1}^A \theta_{1,j}S_1I_j - (\gamma + \mu)I_1 - \alpha_1 I_1 \quad (13)$$

$$\frac{dI_{i \neq 1}}{dt} = \alpha_{i-1}I_{i-1} + \sum_{j=1}^A \theta_{i,j}S_iI_j - (\gamma + \mu)I_i - \alpha_i I_i \quad (14)$$

$$\frac{dR_{i=1}}{dt} = \mu v + \gamma I_1 - \mu R_1 - \alpha_1 R_1 \quad (15)$$

$$\frac{dR_{i \neq 1}}{dt} = \alpha_{i-1}R_{i-1} + \gamma I_i - \mu R_i - \alpha_i R_i \quad (16)$$

where μ is the birth and death rate, α_i is the rate of aging out of age class i which is inversely proportional to its size, γ is the recovery rate, and v is the effective immunization coverage at birth. There is no aging out of the final age class to ensure a constant population scaled to one (i.e., $\alpha_A = 0$).

To calculate $\mathcal{R}_0$ in the age-structured model, we use the Next Generation Matrix method [1]. To do so, we follow the steps from Bjørnstad (2018) [2]:

1. Define the infected states, $\mathbf{X} = (I_1, I_2, \dots, I_A)$

2. Define the new infection terms, $\mathcal{F}(\mathbf{X}) = \left[\sum_{j=1}^A \theta_{i,j} S_i I_j \right]$ for ages $i = 1, 2, \dots, A$

3. Define all losses out of infected states,

$$\mathcal{V}^-(\mathbf{X}) = \begin{bmatrix} (\gamma + \mu + \alpha_1)I_1 \\ \vdots \\ (\gamma + \mu + \alpha_{A-1})I_{A-1} \\ (\gamma + \mu)I_A \end{bmatrix}$$

4. Define all gained transfers between infected states,

$$\mathcal{V}^+(\mathbf{X}) = \begin{bmatrix} 0 \\ \alpha_1 I_1 \\ \vdots \\ \alpha_{A-1} I_{A-1} \end{bmatrix}$$

5. Define the change in infected states,

$$\mathcal{V}(\mathbf{X}) = \mathcal{V}^-(\mathbf{X}) - \mathcal{V}^+(\mathbf{X}) = \begin{bmatrix} (\gamma + \mu + \alpha_1)I_1 \\ (\gamma + \mu + \alpha_2)I_2 - \alpha_1 I_1 \\ \vdots \\ (\gamma + \mu + \alpha_{A-1})I_{A-1} - \alpha_{A-2} I_{A-2} \\ (\gamma + \mu)I_A - \alpha_{A-1} I_{A-1} \end{bmatrix}$$

6. Compute the Jacobians of $\mathcal{F}$ and $\mathcal{V}$. The Jacobian of $\mathcal{V}$ is

$$\mathbf{V} = \left[\frac{\partial \mathcal{V}_i}{\partial X_j} \right] = \begin{bmatrix} \gamma + \mu + \alpha_1 & 0 & 0 & \cdots & 0 & 0 \\ -\alpha_1 & \gamma + \mu + \alpha_2 & 0 & \cdots & 0 & 0 \\ 0 & -\alpha_2 & \gamma + \mu + \alpha_3 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & -\alpha_{A-2} & \gamma + \mu + \alpha_{A-1} & 0 \\ 0 & 0 & \cdots & 0 & -\alpha_{A-1} & \gamma + \mu \end{bmatrix}$$

and the Jacobian of $\mathcal{F}$ is

$$\mathbf{F} = \left[\frac{\partial \mathcal{F}_i}{\partial X_j} \right] = \begin{bmatrix} \theta_{1,1} S_1 & \theta_{1,2} S_1 & \cdots & \theta_{1,A} S_1 \\ \theta_{2,1} S_2 & \theta_{2,2} S_2 & \cdots & \theta_{2,A} S_2 \\ \vdots & \vdots & \ddots & \vdots \\ \theta_{A,1} S_A & \theta_{A,2} S_A & \cdots & \theta_{A,A} S_A \end{bmatrix}$$

7. Then, $\mathcal{R}_0$ is the dominant eigenvalue of the next generation matrix, $\mathbf{K} = \mathbf{FV}^{-1}$ evaluated at the disease-free equilibrium for a particular vaccination level and assuming a stable age distribution. Similarly, $\mathcal{R}_{\text{eff}}(t)$ is the dominant eigenvalue evaluated at time t . We compute these values numerically.

3 The equivalence ratio

[Emily: I've removed the "unity" terminology, as it doesn't seem necessary. The only reason to keep it would be that "unity calculation" is how Ottar describes it in his book, which I think is the (only? or) primary reference/derivation. "Equivalence ratio" is the term I coined for the slightly modified version used in the main text (now described in detail below), but I'm open to other names.]

The dynamics of the age-structured model can be understood in relation to a model of aggregate variables where individuals are assumed to mix homogeneously (i.e., all susceptible individuals mix with all infective and all recovered individuals equally). To equate the dynamics under these two models, incidence can be written as

$$\begin{aligned} \text{age-structured infections} &= \sum_{i=1}^A \sum_{j=1}^A \theta_{i,j} S_i(t) I_j(t) \\ &= \hat{\beta}(t) \sum_{i=1}^A S_i(t) \sum_{j=1}^A I_j(t) \\ &= \text{homogeneous infections} \end{aligned} \tag{17}$$

given the time-varying transmission rate, $\hat{\beta}(t)$, defined as

$$\hat{\beta}(t) = \frac{\sum_{i=1}^A \sum_{j=1}^A \theta_{i,j} S_i(t) I_j(t)}{\sum_{i=1}^A S_i(t) \sum_{j=1}^A I_j(t)}. \tag{18}$$

[Emily: This $\hat{\beta}(t)$ gives us a time-varying reproduction number with dynamics equivalent to an age-structured model. For example, this would reproduce the red line in main text Figure 3C. In the main text, though, we have fixed the WAIFW matrices and then compared the ensuing dynamics (i.e., compared red line to black line). Hopefully my attempt below explains what I've done and the logic.]

In the main text, however, we compared dynamics under structured WAIFW mixing matrices to a flat WAIFW matrix reflective of homogeneous mixing. Let θ and θ^{flat} be the WAIFW matrices that define the transmission dynamics we wish to compare. We choose $\theta_{i,j}^{\text{flat}} = c$ for all i, j such that both models have the same basic reproduction number. Similar to Equation (17), we have

$$\begin{aligned} \text{age-structured infections} &= \sum_{i=1}^A \sum_{j=1}^A \theta_{i,j} S_i(t) I_j(t) \\ \text{flat WAIFW infections} &= \sum_{i=1}^A \sum_{j=1}^A \theta_{i,j}^{\text{flat}} S_i^{\text{flat}}(t) I_j^{\text{flat}}(t) \\ &= c \sum_{i=1}^A S_i^{\text{flat}}(t) \sum_{j=1}^A I_j^{\text{flat}}(t) \end{aligned} \tag{19}$$

which we can use to define an equivalence ratio, $k(t)$, as

$$k(t) = \frac{\sum_{i=1}^A \sum_{j=1}^A \theta_{i,j} S_i(t) I_j(t)}{c \sum_{i=1}^A S_i^{\text{flat}}(t) \sum_{j=1}^A I_j^{\text{flat}}(t)} \quad (20)$$

This ratio can be interpreted as the ratio of the number of new infections in the age-structured model to the number of new infections in the homogeneous model. If $k(t) > 1$ then age-structured dynamics are faster than the homogeneous dynamics, and vice versa if $k(t) < 1$.

[Emily: One challenge with interpretation that I'm not sure how to resolve: because this compares instantaneous incidence, it is sensitive to the relative timing of outbreaks. For this reason, I'm not sure it's very meaningful after the first outbreak. But again, open to thoughts.]

[Emily: I will wait to update the main text until we've verified this calculation and settled on notation/terminology.]

4 Metapopulation model

We consider a simple metapopulation model with P patches that are assumed to be equally connected, such that c is the proportion of contacts that occur within a patch and $1 - c$ is the proportion of contacts that occur between patches. These between-patch contacts are assumed to be distributed uniformly across the remaining $P - 1$ patches, and there is no permanent migration between patches. Patches are assumed to be identical except for a patch-specific effective immunization coverage, v_p . The model is defined by the following system of ordinary differential equations:

$$\frac{dS_p}{dt} = \mu(1 - v_p) - \beta \sum_{q=1}^P C_{p,q} S_p I_q - \mu S_p \quad (21)$$

$$\frac{dI_p}{dt} = \beta \sum_{q=1}^P C_{p,q} S_p I_q - (\gamma + \mu) I_p \quad (22)$$

$$\frac{dR_p}{dt} = \mu v_p + \gamma I_p - \mu R_p \quad (23)$$

where $\mathbf{C}$ is the connectivity matrix, with $C_{p,p} = c$ for all $p = 1, \dots, P$ and $C_{p,q} = (1 - c)/(P - 1)$ for $p \neq q$. Again, we use the Next Generation Matrix method [1] to calculate $\mathcal{R}_0$ following the steps from Bjørnstad (2018) [2]:

1. Define the infected states, $\mathbf{X} = (I_1, I_2, \dots, I_P)$
2. Define the new infection terms, $\mathcal{F}(\mathbf{X}) = \left[\beta \sum_{q=1}^P C_{p,q} S_p I_q \right]$ for patches $p = 1, 2, \dots, P$
3. Define all losses out of infected states,

$$\mathcal{V}^-(\mathbf{X}) = \begin{bmatrix} (\gamma + \mu) I_1 \\ \vdots \\ (\gamma + \mu) I_P \end{bmatrix}$$

138 4. Define all gained transfers between infected states, $\mathcal{V}^+(\mathbf{X}) = 0$

139 5. Define the change in infected states,

$$\mathcal{V}(\mathbf{X}) = \mathcal{V}^-(\mathbf{X}) - \mathcal{V}^+(\mathbf{X}) = \begin{bmatrix} (\gamma + \mu)I_1 \\ \vdots \\ (\gamma + \mu)I_P \end{bmatrix}$$

140 6. Compute the Jacobians of $\mathcal{F}$ and $\mathcal{V}$. The Jacobian of $\mathcal{V}$ is a diagonal matrix,
141 $\mathbf{V} = \text{diag}(\gamma + \mu)$, and thus the inverse $\mathbf{V}^{-1} = \text{diag}\left(\frac{1}{\gamma + \mu}\right)$. The Jacobian of $\mathcal{F}$ is

$$\mathbf{F} = \left[\frac{\partial \mathcal{F}_p}{\partial X_q} \right] = \beta \begin{bmatrix} C_{1,1}S_1 & C_{1,2}S_1 & \cdots & C_{1,P}S_1 \\ C_{2,1}S_2 & C_{2,2}S_2 & \cdots & C_{2,P}S_2 \\ \vdots & \vdots & \ddots & \vdots \\ C_{P,1}S_P & C_{P,2}S_P & \cdots & C_{P,P}S_P \end{bmatrix}$$

142 7. This yields the next generation matrix, $\mathbf{K} = \mathbf{FV}^{-1} = \frac{\beta}{\gamma + \mu} \text{diag}(\mathbf{S})\mathbf{C}$ where $\mathbf{S} =$
143 $(S_1, S_2, \dots, S_P)$. We evaluate $\mathbf{K}$ at the disease-free equilibrium or at time t and
144 calculate the dominant eigenvalue numerically to obtain $\mathcal{R}_0$ and $\mathcal{R}_{\text{eff}}$, respectively.

5 Supplementary Figures

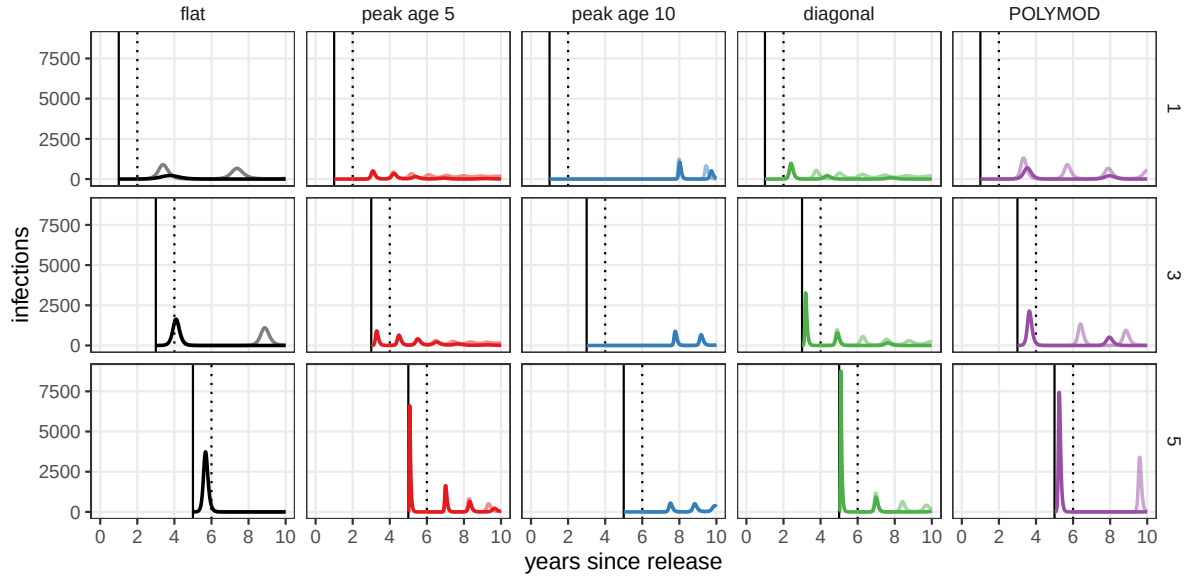


Figure S1: Sensitivity of results to the timing of infection introduction. For each WAIFW matrix (columns, see main text Figure 3A), simulated dynamics when a single infection is introduced into the 5 year old age class 1, 3, or 5 years after the drop in effective immunization coverage (rows). The solid vertical line shows the timing of introduction. The lighter line shows simulation results if effective immunization coverage returns to its pre-decline level 1 year after the drop (shown by dotted line).

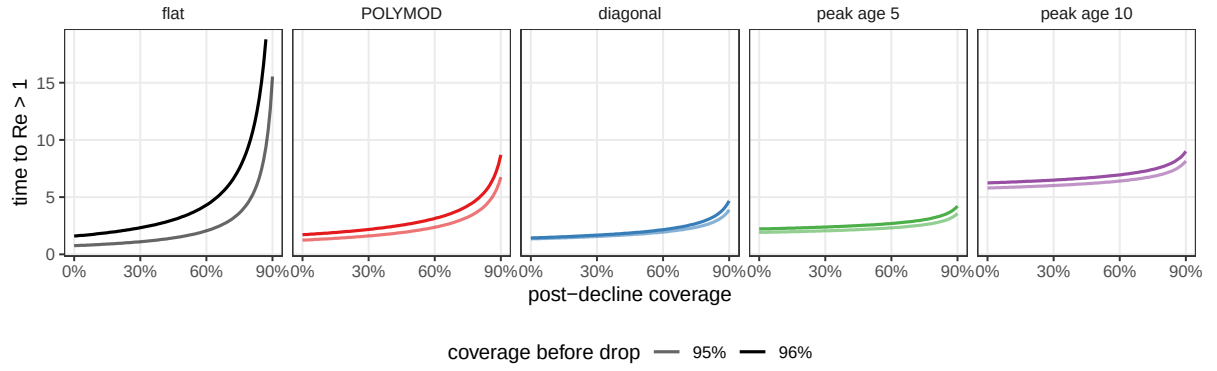


Figure S2: Theoretical honeymoon time predictions, computed as time until $\mathcal{R}_{\text{eff}} > 1$, under five assumptions about age-structured mixing. Results are shown for a population with $\mu = 1/80 \text{ years}^{-1}$, a pathogen with $\mathcal{R}_0 = 17$, and pre-decline effective immunization coverage of 95% and 96%.

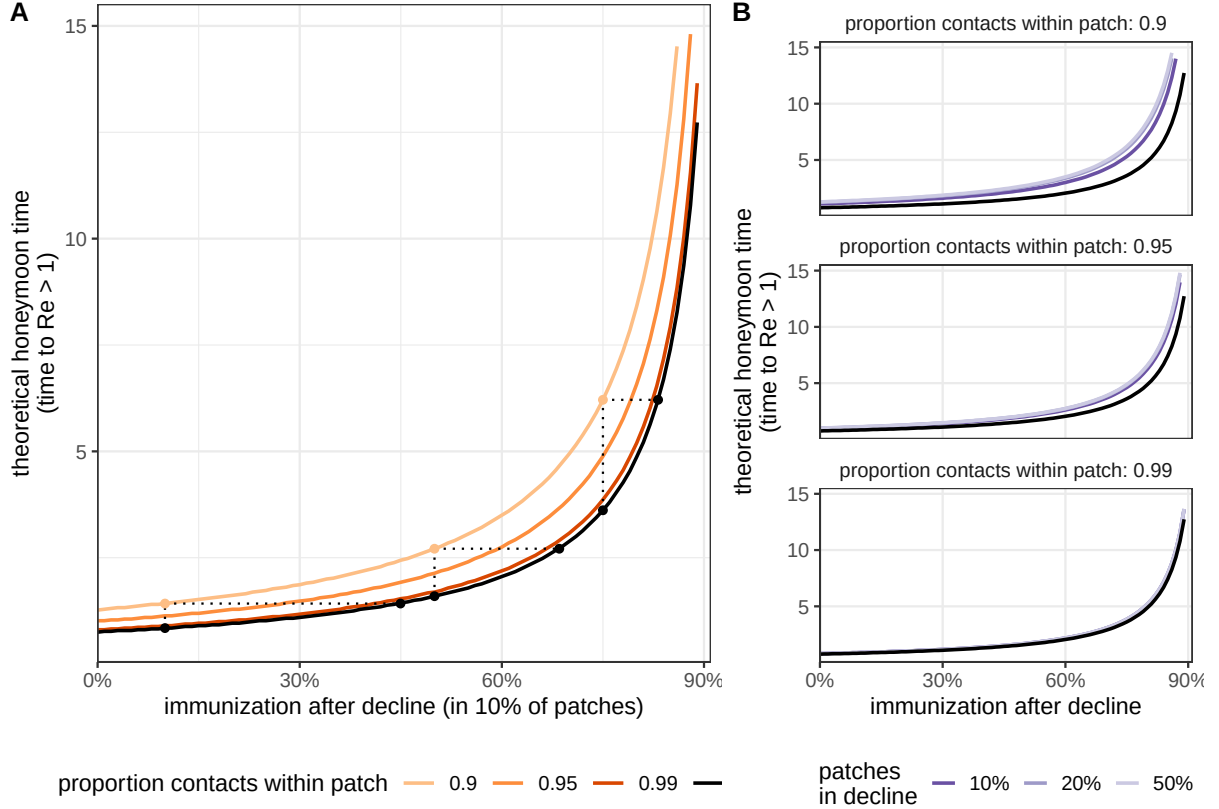


Figure S3: Role of patch connectivity in theoretical honeymoon time. (A) Theoretical honeymoon time, computed as time until $\mathcal{R}_{\text{eff}} > 1$, by post-decline effective immunization coverage when drops occur in 10% of patches among an equally connected network. Line color shows different connectivity scenarios, modeled by the proportion of contacts that occur within the home patch, and the points show equivalence between a model with no connectivity (i.e., 100% of contacts occurring within the home patch) and a model where 90% of contacts occur within the home patch. (B) Sensitivity of results to the percent of patches in which effective immunization coverage declines compared to the no connectivity scenario (black). Results are shown for a population with $\mu = 1/80 \text{ years}^{-1}$, a pathogen with $\mathcal{R}_0 = 17$, and pre-decline effective immunization coverage of 95%.

References

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- [2] Bjørnstad ON. *Epidemics: Models and Data using R. Use R!*. Springer International Publishing; 2018. doi:10.1007/978-3-319-97487-3.